178037: biotite granodiorite, Nelson Well

Location and sampling

MARBLE BAR (SF 50-8), MARBLE BAR (2855)
MGA Zone 50, 799430E 7649660N

Sampled on 6 October 2002

The sample was taken from a 1 m-high, 3 m-long whaleback in a small cluster of pavements and boulders located 250 m southwest of the access track and 3 km west-northwest of Nelson Well.

Tectonic unit/relations

The sample is a pale grey, strongly foliated, medium- and even-grained biotite granodiorite from the Mount Edgar Granitoid Complex, East Pilbara Granite–Greenstone Terrane (Hickman and Van Kranendonk, in prep.). At the sampling site, the monzogranite has been intruded by leucocratic granitic dykes and veins that are up to 3 cm thick and aligned with the foliation. The sample was free of obvious veins. The granodiorite forms a major intrusive unit in the southeastern part of the complex, and its crystallization age and relationships with other intrusive suites must be established for regional interpretations.

Petrographic description

The principal minerals in this sample are plagioclase (45–50 vol.%), quartz (35 vol.%), K-feldspar (15 vol.%), and biotite (2–3 vol.%), with trace amounts of accessory opaque oxide, titanite, muscovite, epidote, prehnite, limonite, and zircon. Plagioclase forms rounded anhedral grains from 1 to 5 mm in length, with relatively fine grained anhedral microcline interstitial to or rimming the plagioclase. There is minor myrmekite between plagioclase and microcline. Small lenses of fine-grained feldspars, usually with minor myrmekite, are scattered and have recrystallized. In some areas, these aggregates also contain minor quartz, but most of the quartz is between 0.5 and 3 mm in grain size, with subgrains mostly in the larger grains. The minor biotite, up to 1.5 mm in grain size, is mostly foliated, with some areas suggesting an S–C fabric. Very minor opaque oxide and titanite are present, suggesting some oxidized I-type characteristics. The rare zircon crystals are mostly less than 40 µm long, and are clear and unaltered. The plagioclase is partly fresh and probably andesine–oligoclase, but has irregularly developed alteration to sericite with or without clinozoisite, locally intense in areas of high fluid activity, with sparse poikilitic flakes of muscovite or granular epidote in some patches, of secondary origin. Much of the biotite is altered to chlorite with or without leucoxene, with epidote in some grains, usually carrying lamellae of leucoxene. There are patches of prehnite in other grains, with clay alteration or patches of probable ferristilpnomelane in places. The biotite alteration represents a range of temperatures from greenschist facies to possible prehnite–pumpellyite facies. Minor ferristilpnomelane has also developed adjacent to the opaque oxide. Traces of limonite could replace microcrystalline pyrite. The textures in this sample suggest some low amphibolite-facies recrystallization and deformation as well as gradual hydrothermal alteration at prehnite–pumpellyite- to greenschist-facies temperatures of 250–400°C.

Zircon morphology

The zircons isolated from this sample are typically yellowish-brown or dark brown, between 30 × 100 µm and 140 × 300 µm in size, and elongate and euhedral. Most have euheral zonation but a minority have narrow, cylindrical hollow cores or are structureless and lack internal igneous zonation. Many grains are metamict. Cathodoluminescence images of representative zircons are given in Figure 1

Analytical details

This sample was analysed on 8 September 2003. The counter deadtime during the analysis session was 24 ns. Six analyses of the CZ3 standard obtained during the analysis session indicated a Pb*/U calibration uncertainty of 1.16% (1σ). Common-Pb corrections were applied assuming Broken Hill common-Pb isotopic compositions for all analyses, with the exception of analyses 11.1, 13.1, 14.1, 17.1, and 18.1, for which isotopic compositions determined using the method of Cumming and Richards (1975) were assumed.

Results

Twenty-one analyses were obtained from 21 zircons. Results are given in Table 1 and shown on a concordia plot in Figure 2.

Interpretation

The analyses are concordant to slightly discordant, with the discordance pattern consistent with several episodes, including a recent episode, of radiogenic-Pb loss. Fifteen concordant and highly discordant analyses of 15 zircons have $^{207}Pb/^{206}Pb$ ratios defining a single population and indicating a weighted mean $^{207}Pb/^{206}Pb$ date of 3313 ± 2 Ma (chi-squared = 1.61). The remaining analyses are highly discordant and cannot be confidently grouped.

The date of 3313 ± 2 Ma indicated by the weighted mean $^{207}Pb/^{206}Pb$ ratio of 15 concordant and highly discordant analyses of 15 zircons is interpreted as the

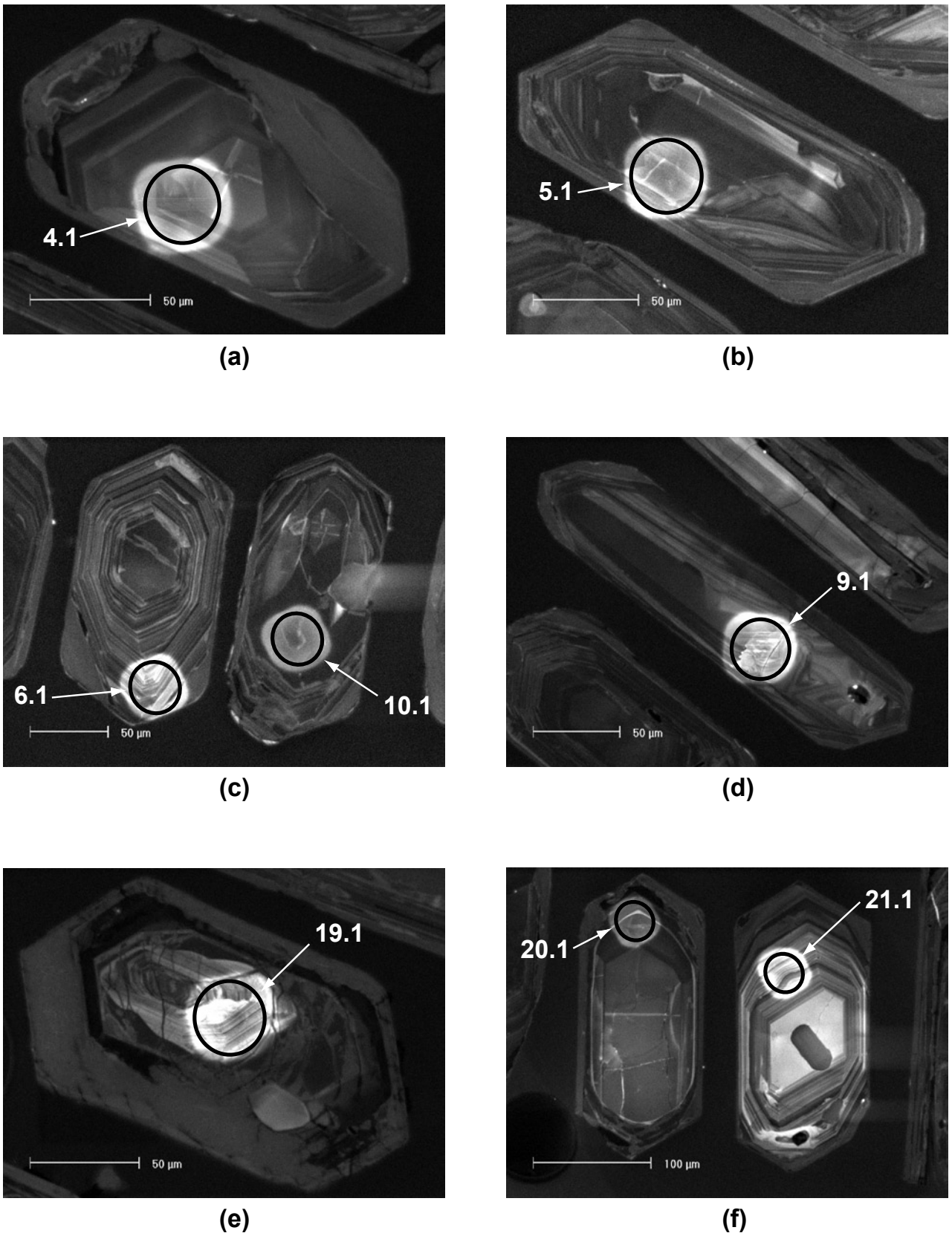

Figure 1. Cathodoluminescence images of representative zircons from sample 178037: biotite granodiorite, Nelson Well

Table 1. Ion microprobe analytical results for zircons from sample 178037: biotite granodiorite, Nelson Well

Grain .spot	*U (ppm)*	*Th (ppm)*	*Pb (ppm)*	*f206%*	$^{207}Pb/^{206}Pb$	±1σ	$^{208}Pb/^{206}Pb$	±1σ	$^{206}Pb/^{238}U$	±1σ	$^{207}Pb/^{235}U$	±1σ	*% concordance*	$^{207}Pb/^{206}Pb$ *Age*	±1σ
1.1	532	175	321	0.137	0.25860	0.00042	0.08773	0.00043	0.5191	0.0061	18.509	0.222	83	3 238	3
2.1	304	92	232	0.189	0.27213	0.00052	0.08344	0.00056	0.6501	0.0077	24.391	0.297	97	3 318	3
3.1	255	134	194	0.076	0.26122	0.00052	0.14824	0.00057	0.6262	0.0074	22.554	0.275	96	3 254	3
4.1	242	132	198	0.051	0.27189	0.00055	0.14370	0.00057	0.6707	0.0079	25.145	0.308	100	3 317	3
5.1	230	86	181	0.096	0.27117	0.00057	0.10032	0.00056	0.6659	0.0079	24.898	0.306	99	3 313	3
6.1	309	124	235	0.040	0.27002	0.00048	0.10875	0.00045	0.6415	0.0076	23.883	0.290	97	3 306	3
7.1	132	44	101	0.142	0.27005	0.00078	0.08807	0.00082	0.6536	0.0079	24.337	0.310	98	3 306	5
8.1	229	62	176	0.184	0.27120	0.00056	0.07204	0.00058	0.6602	0.0078	24.687	0.302	99	3 313	3
9.1	111	25	86	0.157	0.27089	0.00086	0.05976	0.00089	0.6711	0.0081	25.065	0.323	100	3 311	5
10.1	450	165	352	0.016	0.27150	0.00040	0.09662	0.00032	0.6647	0.0078	24.883	0.298	99	3 315	2
11.1	248	124	201	0.703	0.27097	0.00065	0.13516	0.00100	0.6541	0.0077	24.438	0.302	98	3 312	4
12.1	484	250	390	0.006	0.27112	0.00039	0.13797	0.00039	0.6635	0.0078	24.804	0.297	99	3 313	2
13.1	726	241	508	0.078	0.26861	0.00031	0.08421	0.00028	0.5985	0.0070	22.166	0.263	92	3 298	2
14.1	396	60	299	0.147	0.27157	0.00043	0.03987	0.00037	0.6665	0.0078	24.958	0.300	99	3 315	2
15.1	246	96	187	0.203	0.26960	0.00058	0.10079	0.00064	0.6416	0.0076	23.849	0.293	97	3 304	3
16.1	381	93	288	0.100	0.27048	0.00046	0.06732	0.00039	0.6530	0.0077	24.352	0.294	98	3 309	3
17.1	887	338	528	0.089	0.26136	0.00031	0.09352	0.00029	0.5091	0.0059	18.346	0.217	82	3 255	2
18.1	653	161	367	0.163	0.25597	0.00038	0.07964	0.00040	0.4865	0.0057	17.169	0.205	79	3 222	2
19.1	85	30	62	0.131	0.26873	0.00099	0.09638	0.00109	0.6161	0.0076	22.827	0.302	94	3 299	6
20.1	302	148	244	0.048	0.27159	0.00048	0.13101	0.00048	0.6678	0.0079	25.007	0.304	99	3 315	3
21.1	88	33	70	0.104	0.27149	0.00096	0.10119	0.00101	0.6753	0.0083	25.279	0.333	100	3 315	6

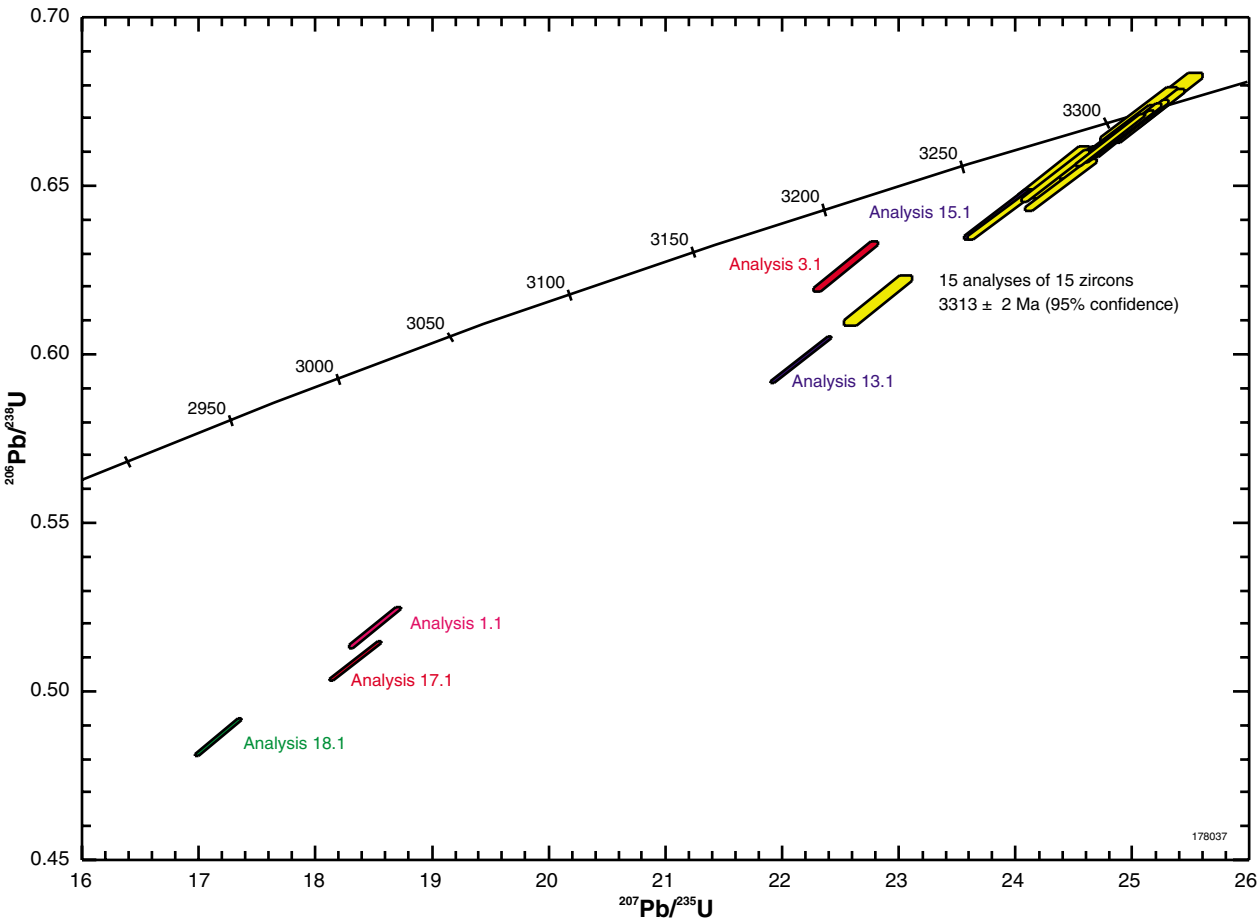

Figure 2. Concordia plot for sample 178037: biotite granodiorite, Nelson Well

age of igneous crystallization of the granodiorite. The remaining discordant analyses are interpreted to be of sites that have lost radiogenic Pb.

Recommended reference for this publication:

NELSON, D. R., 2005, 178037: biotite granodiorite, Nelson Well; Geochronology dataset 561; in Compilation of geochronology data, June 2006 update: Western Australia Geological Survey.

Data obtained: 08/09/2003; Data released: 30/06/2005